Trigonometry: Harmonic Forms

Questions 1–20

Question 1. Each of the following can be written in at least one of the forms $R\sin(\theta \pm \alpha)$ or $R\cos(\theta \pm \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. Write down the exact value of R in each case:

a) $8\sin\theta + 15\cos\theta$
b) $12\sin\theta - 35\cos\theta$
c) $14\cos\theta - 20\sin\theta$
d) $2\sqrt{2}\cos\theta + 5\sqrt{2}\sin\theta$

Question 2. State the maximum and minimum values of each of the following expressions:

a) $9\sin\theta - 40\cos\theta$
b) $10\sin\theta + 24\cos\theta$
c) $6\sqrt{3}\cos\theta + 4\sqrt{3}\sin\theta$
d) $3\sqrt{5}\cos\theta - 5\sqrt{3}\sin\theta$

Question 3. State the maximum and minimum values of each of the following expressions:

a) $12 + 220\cos\theta - 21\sin\theta$

b) $50 - \cos\theta + 18\sin\theta$

c) $(16\cos\theta + 63\sin\theta)^2$

d) $(5 + 4\sin\theta - 9\cos\theta)^2$

Question 4. State the maximum and minimum values of each of the following expressions:

a) $\dfrac{1}{14 + 5\sin\theta + 12\cos\theta}$

b) $\dfrac{15}{90 - 84\sin\theta - 13\cos\theta}$

c) $\dfrac{100}{(6 + \sin\theta + \cos\theta)^2}$

d) $\dfrac{625}{90 + (-33\cos\theta + 56\sin\theta)^2}$

Question 5. The maximum value of the expression $\sqrt{10 + 3\sqrt{3}\cos\theta + k\sin\theta}$ is 4. Find the possible values of k.

Question 6. Write the following in the form $R\sin(x + \alpha)$ where $R > 0$ and $0° < \alpha < 90°$.

a) $3\sin x + 3\cos x$
b) $5\sin x + 5\sqrt{3}\cos x$
c) $9\sqrt{3}\sin x + 9\cos x$
d) $4\sqrt{2}\sin x + 2\sqrt{2}\cos x$

Question 7. Write the following in the form $R\sin(x-\alpha)$ where $R>0$ and $0<\alpha<\frac{\pi}{2}$.

a) $2\sqrt{2}\sin x - 2\sqrt{2}\cos x$
b) $3\sin x - 4\cos x$
c) $12\sin x - 5\cos x$
d) $11\sin x - \cos x$

Question 8. Write the following in the form $R\cos(x+\alpha)$ where $R>0$ and $0^\circ<\alpha<90^\circ$.

a) $4\cos x - 4\sqrt{3}\sin x$
b) $5\cos x - \sin x$
c) $20\cos x - 30\sin x$
d) $13\cos x - 6\sin x$

Question 9. Write the following in the form $R\cos(x-\alpha)$ where $R>0$ and $0<\alpha<\frac{\pi}{2}$.

a) $20\cos x+99\sin x$
b) $56\cos x+33\sin x$
c) $25\sqrt{3}\cos x+75\sin x$
d) $2\sqrt{5}\cos x+5\sqrt{5}\sin x$

Question 10. Write each of the following in an appropriate harmonic form: $R\sin(\theta\pm\alpha)$ or $R\cos(\theta\pm\alpha)$, where $R>0$ and $0^\circ<\alpha<90^\circ$.

a) $6\cos\theta-8\sin\theta$
b) $7\cos\theta+24\sin\theta$
c) $5\sin\theta-5\cos\theta$
d) $8\sqrt{2}\sin\theta+10\sqrt{2}\cos\theta$

Question 11. Find the maximum value of the following expressions, and the smallest positive value of θ, in degrees, for which this maximum occurs:

a) $\sin\theta + 5\cos\theta$
b) $9\cos\theta - \sqrt{2}\sin\theta$
c) $6\sin\theta - 4\cos\theta$
d) $10\cos\theta + 20\sin\theta$

Question 12. Solve the following trigonometric equations:

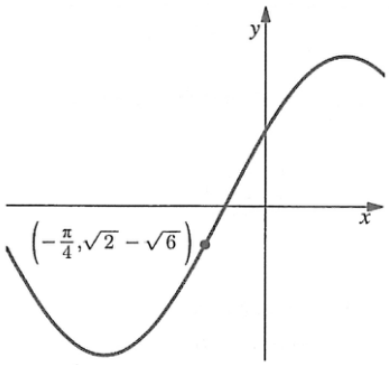
a) $4\sqrt{2}\sin\theta - 4\sqrt{2}\cos\theta = 7,\ 0° < \theta < 360°$
b) $4\sin\theta + 2\cos\theta = 3,\ 0 < \theta < 2\pi$
c) $6\sqrt{3}\cos 3\theta + 5\sin 3\theta = 1,\ 0° < \theta < 240°$
d) $10\sqrt{5}\cos 2\theta - 12\sqrt{5}\sin 2\theta = 5,\ 0 < \theta < 2\pi$

Question 13. Describe a sequence of two graph transformations, in radians, that would map $y = \sin x$ onto the image $y = 4 + 10\sin x - \sqrt{2}\cos x$.

Question 14. Write $40\cos 5x - 42\sin 5x$ in the form $R\cos(5x + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

Question 15. Write $8\cos\frac{1}{2}x + 3\sin\frac{1}{2}x$ in the form $R\cos(\frac{1}{2}x - \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

Question 16. The diagram below shows a curve with equation $y = p\sin x + q\cos x$, where p and q are positive constants. The curve passes through the point $(-\frac{\pi}{4}, \sqrt{2} - \sqrt{6})$. It is given that the minimum value of the curve is -4. Find the exact values of p and q.



Question 17. For a land-based wind turbine, the height above the ground, h metres, of the tip of one of the wind turbine blades after t seconds is modelled by the equation

$$h = k + 28\cos 4t + 54\sin 4t.$$

a) Before the wind turbine is switched on and starts moving, the blade tip in question is locked into a position 117 metres above the ground. Find the value of k.
b) The wind turbine is switched on. Find the maximum height the blade tip reaches above the ground as it rotates, and the time it takes the blade tip to first reach this position.

Question 18. The depth of water, D meters, at a point in a tidal river is modelled by the equation

$$D = 4 + \sin(\tfrac{\pi}{6}t - 2) + 1.5\cos(\tfrac{\pi}{6}t - 2)$$

where t is the number of hours after midnight.

a) Find the minimum depth of the river at the point being measured.
b) A ferry can only cross the river when the depth of the water is at least 5 metres. Derek arrives at the ferry at 1pm. How long must he wait before the ferry starts operating again?

Question 19. An engine piston moves up and down in a vertical cylinder. The point A is at the top of the piston. The vertical displacement, h mm, of A from its lowest position is modelled by the equation

$$h = 50 + 40\cos 20t - 30\sin 20t$$

where t is the time in seconds after the piston starts moving.

a) Find the maximum displacement of A.

b) In a single cycle of the engine, A is more than 80 millimetres above its lowest position for T seconds. Find the value of T.

c) The engine's speed is increased so that the piston now completes 15 cycles per second. Explain how the equation of the model should be modified to reflect this change in speed.

Question 20.

a) Express $\sqrt{129}\cos\theta - 20\sin\theta$ in the form $a\cos(\theta + b)$, where $a > 0$ and $0 < b < 90°$.

A meteorologist studies the temperature and humidity in Saskatoon over one year. The temperature, $T°$C, is modelled by the equation

$$T = \sqrt{129}\cos(30t)° - 20\sin(30t)° + C$$

where t is the time in months after the start of the year.

a) Given that during the year, the maximum temperature in Saskatoon is 25°C, find the value of C.

b) The humidity, $H\%$, during the same year is modelled by $H = 10\cos(30t + 45)° + 72$. Find the temperature in Saskatoon when the humidity is at its maximum.